

End To End Retail Data Pipeline

TEAM MEMBERS:-

Anurag Sarkar
Avranil Set
Md.Wasim



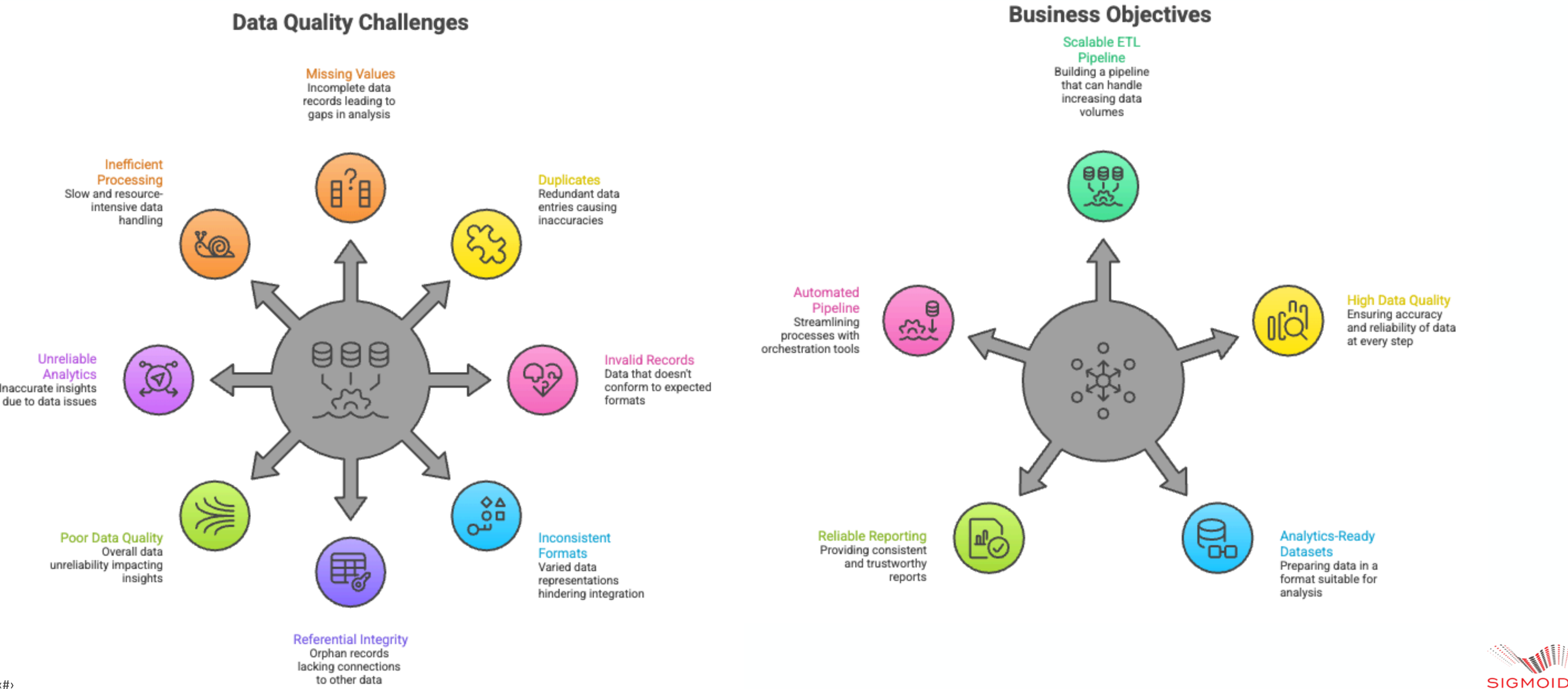
Agenda

1. Problem Statement
2. Tech Stack Overview
3. System Architecture (Medallion Design)
4. Data Sources & Dataset Generation
5. Bronze Layer (Ingestion)
6. Silver Layer (Cleaning & Validation)
7. Gold Layer (Data Modeling)
8. Data Quality Framework
9. Orchestration using Airflow
10. Snowflake Integration
11. Business Insight
12. What We Achieved
13. Challenges & Engineering Decisions

Problem Statement & Business Objectives

Overview

- We were tasked with building a production-grade retail data pipeline that processes raw CSV data into analytics-ready datasets.
- Ensure data reliability, scalability, and quality using modern data engineering practices.
- Enable business insights through structured data modeling and warehousing.



Tech Stack

Processing Engine



Databricks



Apache Spark

Programming



Python



SQL

Data Warehouse



Snowflake

Orchestration



Airflow



Docker

Version Control

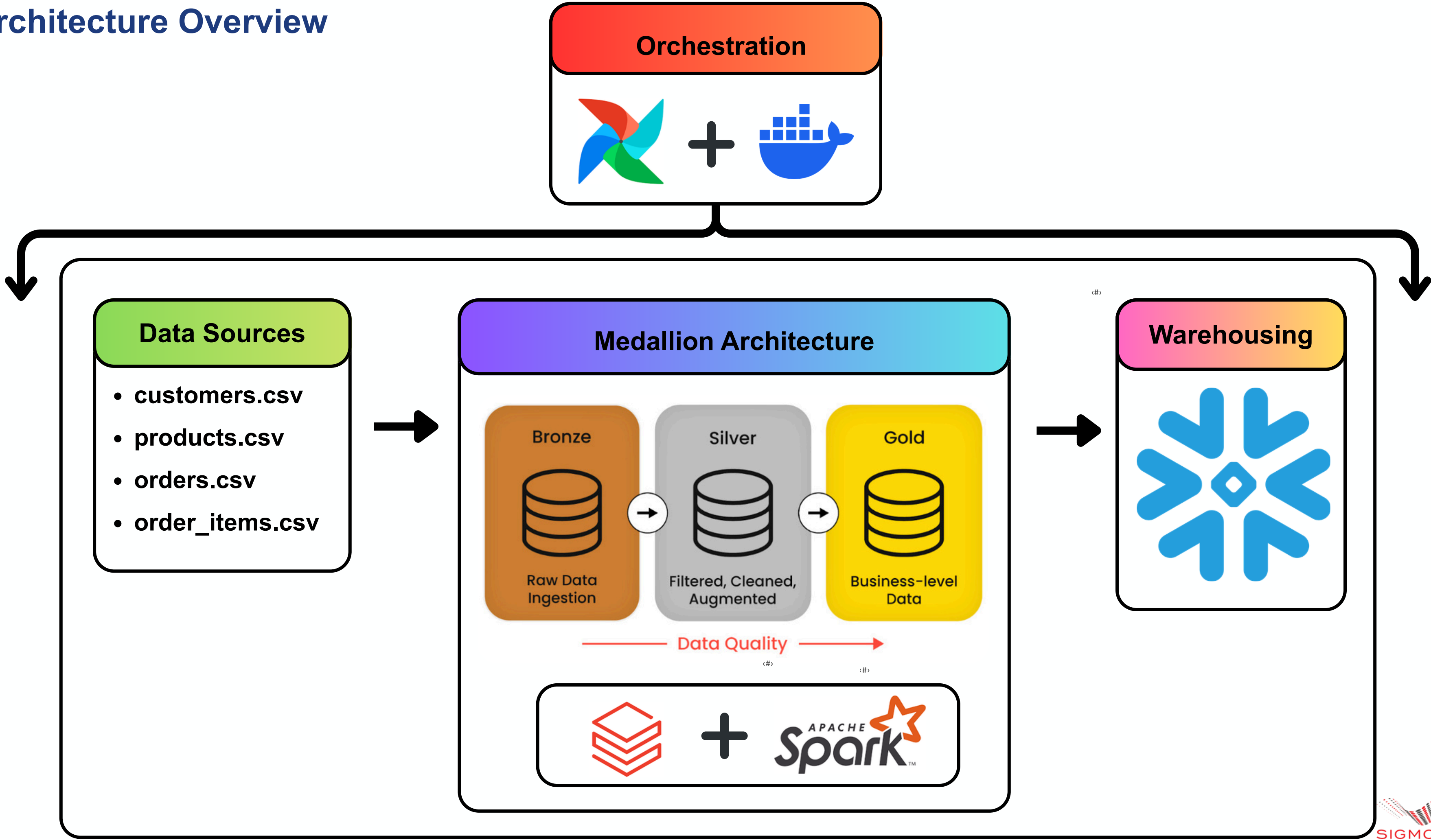


Git



GitHub

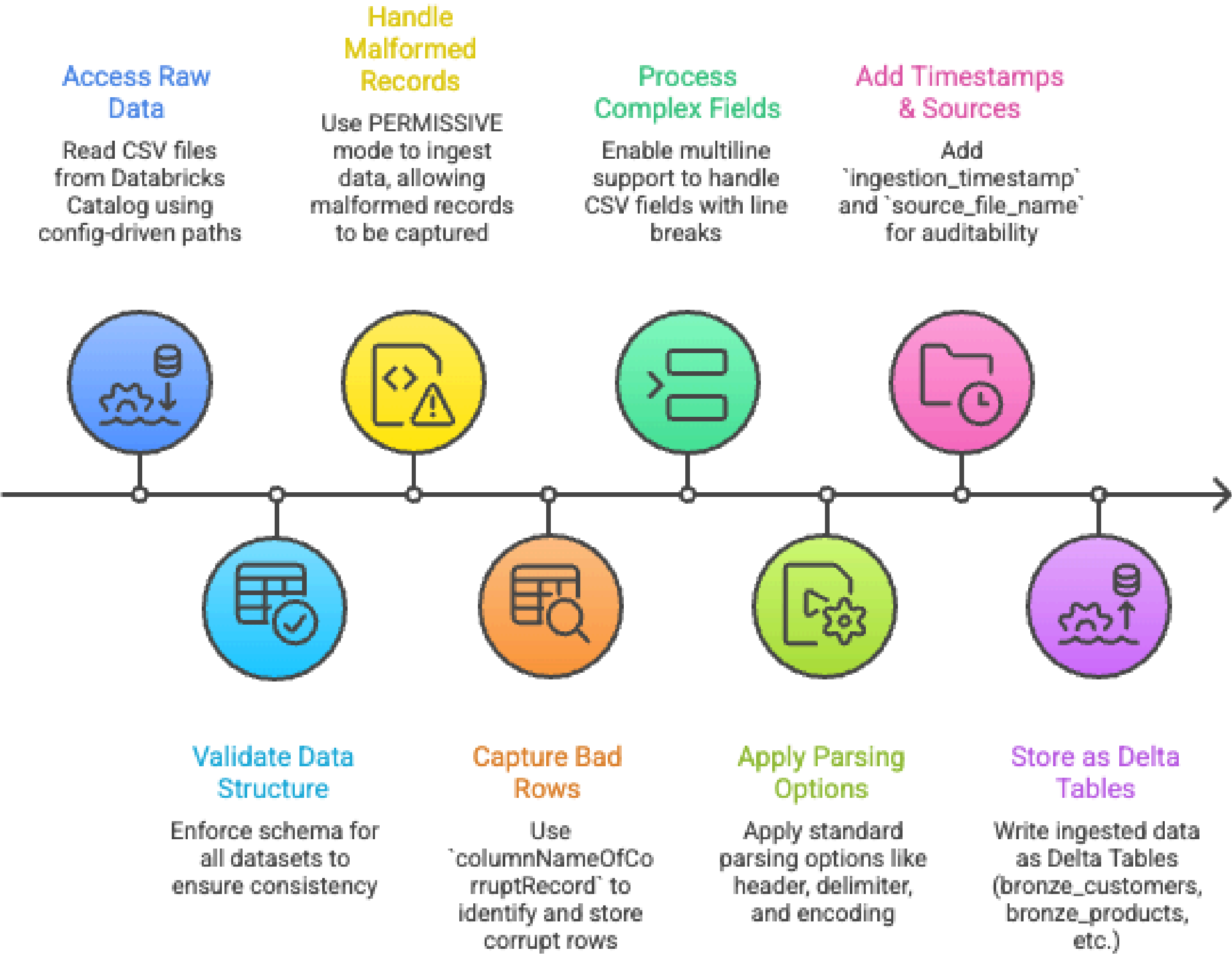
Architecture Overview



Data Sources

Source	Format	Total Rows	What it tells us
customers.csv	CSV	~2k	Customer details
products.csv	CSV	~150	Product catalog
orders.csv	CSV	~5k	Order records
order_items.csv	CSV	~12k	Order line items

Bronze Layer (Ingestion)



Bronze Layer Tables

Catalog Explorer > capstone_project_catalog > bronze >

bronze_customers

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OverviewSample DataDetailsPermissionsPoliciesHistoryLineageInsightsQuality

Ask your question about the sample data...NewAsk

How many customers signed up per day over time?Which countries have the most customer signups?Are there duplicate email addresses across custo

Sample

	customer_id	first_name	last_name	email	signup_date	country	ingestion_times
1	984	Patricia	Davis	pdavis@yandex.com	2022-05-01	United States	2026-04-17T14:44:3
2	1597	Oliver	Bhat	oliver.bhat@protonmail.com	2024-01-11	United States	2026-04-17T14:44:3
3	1906	Lisa	Moore	lisa.moore@gmail.com	2020-03-06	United States	2026-04-17T14:44:3
4	1708	Priya	Anderson	priya.anderson60@zoho.com	2020-03-10	United States	2026-04-17T14:44:3
5	1012	Benjamin	Iyer	benjamin.iyer26@zoho.com	2020-05-05	United States	2026-04-17T14:44:3
6	846	Karen	Martinez	karen_martinez@zoho.com	2022-02-08	India	2026-04-17T14:44:3
7	620	Oliver	Desai	oliver.desai@zoho.com	2022-04-21	United States	2026-04-17T14:44:3
8	269	Liam	Patel	liampat@hotmail.com	2022-02-07	India	2026-04-17T14:44:3
9	648	Emma	White	emma.white@gmail.com	2024-06-21	Australia	2026-04-17T14:44:3
10	161	Ella	Bhat	ebhat@mail.com	2022-05-24	India	2026-04-17T14:44:3
11	1304	Oliver	Kumar	oliver_kumar@nmail.com	2023-02-18	Janan	2026-04-17T14:44:3

Customers Table

Catalog Explorer > capstone_project_catalog > bronze >

bronze_order_items

Open in a dashboardShare

OverviewSample DataDetailsPermissionsPoliciesHistoryLineageInsightsQuality

Ask your question about the sample data...NewAsk

Are there duplicate order_item_ids in the table?How does average quantity per order vary over time?Which source_file_name contribut

Sample

	order_item_id	order_id	product_id	quantity	ingestion_timestamp	source_file_name
1	3972	1787	134	2	2026-04-17T14:44:45.683+00:...	order_items.csv
2	10615	4773	131	1	2026-04-17T14:44:45.683+00:...	order_items.csv
3	9402	4209	17	3	2026-04-17T14:44:45.683+00:...	order_items.csv
4	3929	1767	59	2	2026-04-17T14:44:45.683+00:...	order_items.csv
5	6719	3026	31	1	2026-04-17T14:44:45.683+00:...	order_items.csv
6	10525	4725	66	2	2026-04-17T14:44:45.683+00:...	order_items.csv
7	9796	4392	88	4	2026-04-17T14:44:45.683+00:...	order_items.csv
8	7500	3359	136	2	2026-04-17T14:44:45.683+00:...	order_items.csv
9	10252	4593	3	1	2026-04-17T14:44:45.683+00:...	order_items.csv
10	9954	4462	87	1	2026-04-17T14:44:45.683+00:...	order_items.csv
11	9183	4106	5	4	2026-04-17T14:44:45.683+00:...	order_items.csv
12	9315	4162	85	8	2026-04-17T14:44:45.683+00:...	order_items.csv

Order Items Table

Catalog Explorer > capstone_project_catalog > bronze >

bronze_products

Open in a dashboardShare

OverviewSample DataDetailsPermissionsPoliciesHistoryLineageInsightsQuality

Ask your question about the sample data...NewAsk

Which product categories have the highest average prices?Are there products with missing or zero prices?How many products were ingeste

Sample

	product_id	product_name	category	price	ingestion_timestamp	source_file_name
1	1	Pro Mouse	Electronics	77.44	2026-04-17T14:44:41.438+00:...	products.csv
2	2	HD Monitor Light	Electronics	36.42	2026-04-17T14:44:41.438+00:...	products.csv
3	3	Bluetooth Headphones	Electronics	225.44	2026-04-17T14:44:41.438+00:...	products.csv
4	4	Bluetooth Mouse	Electronics	263.59	2026-04-17T14:44:41.438+00:...	products.csv
5	5	Pro Camera	Electronics	285.31	2026-04-17T14:44:41.438+00:...	products.csv
6	6	Pro Speaker	Electronics	159.32	2026-04-17T14:44:41.438+00:...	products.csv
7	7	HD Speaker	Electronics	44.88	2026-04-17T14:44:41.438+00:...	products.csv
8	8	Wireless Monitor Light	Electronics	268.61	2026-04-17T14:44:41.438+00:...	products.csv
9	9	Smart Headphones	Electronics	10.22	2026-04-17T14:44:41.438+00:...	products.csv
10	10	Wireless Charger	Electronics	101.16	2026-04-17T14:44:41.438+00:...	products.csv
11	11	Pro Camera	Electronics	131.8	2026-04-17T14:44:41.438+00:...	products.csv

Products Table

Catalog Explorer > capstone_project_catalog > bronze >

bronze_orders

Open in a dashboardShare

OverviewSample DataDetailsPermissionsPoliciesHistoryLineageInsightsQuality

Ask your question about the sample data...NewAsk

How many orders exist for each distinct status value?What is the daily order volume trend over time?Which source files contributed

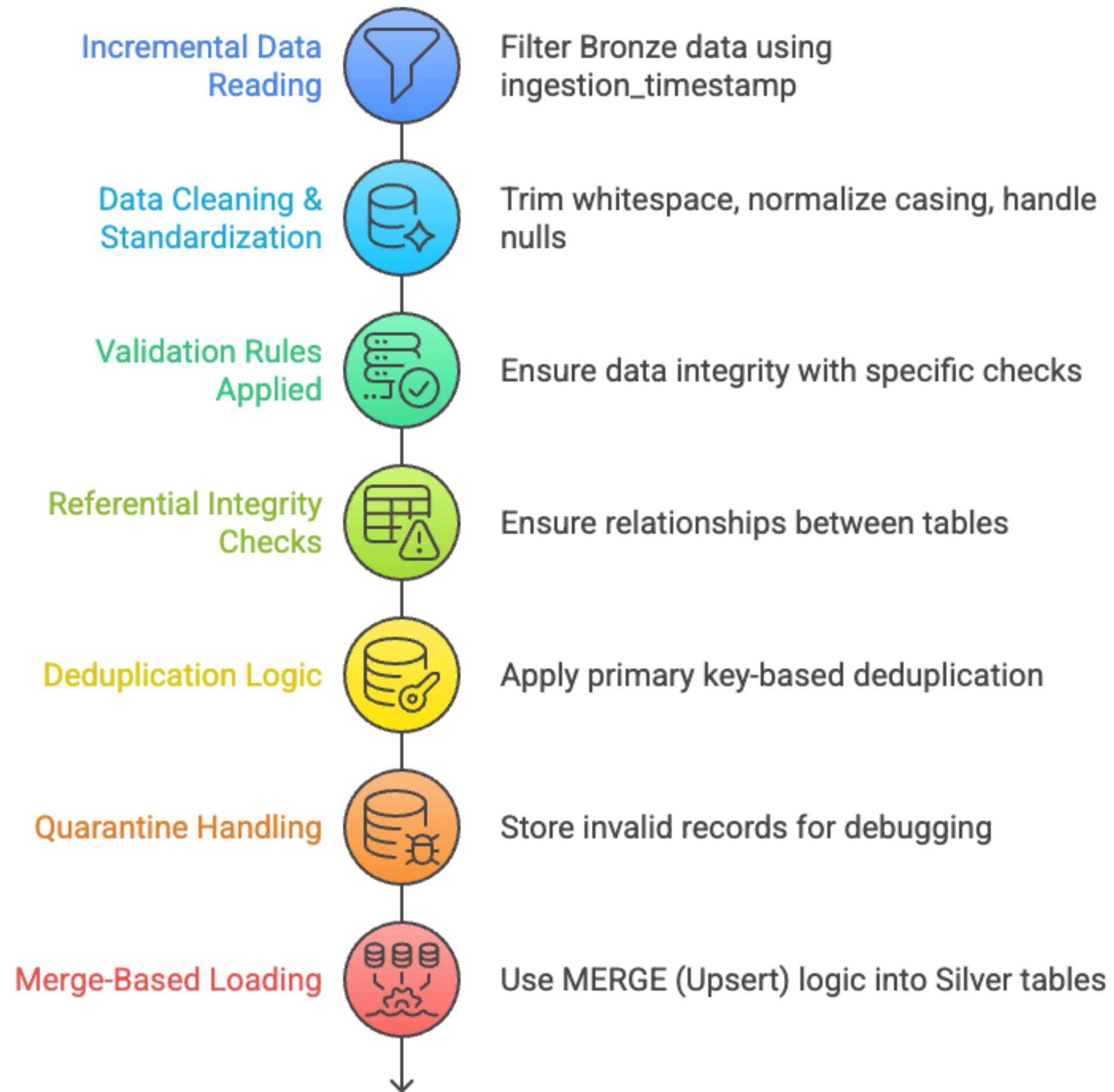
Sample

	order_id	customer_id	order_date	status	ingestion_timestamp	source_file_name
3	3	67	2024-11-17	CREATED	2026-04-17T14:44:43.634+00:...	orders.csv
4	4	264	2022-09-21	DELIVERED	2026-04-17T14:44:43.634+00:...	orders.csv
5	5	857	2022-01-22	CANCELLED	2026-04-17T14:44:43.634+00:...	orders.csv
6	6	1528	2024-12-19	SHIPPED	2026-04-17T14:44:43.634+00:...	orders.csv
7	7	292	2023-07-17	SHIPPED	2026-04-17T14:44:43.634+00:...	orders.csv
8	8	891	2025-03-19	CREATED	2026-04-17T14:44:43.634+00:...	orders.csv
9	9	1785	2024-05-01	DELIVERED	2026-04-17T14:44:43.634+00:...	orders.csv
10	10	874	2024-02-29	RETURNED	2026-04-17T14:44:43.634+00:...	orders.csv
11	11	1758	2022-01-03	SHIPPED	2026-04-17T14:44:43.634+00:...	orders.csv
12	12	798	2022-07-23	DELIVERED	2026-04-17T14:44:43.634+00:...	orders.csv
13	13	1120	2023-08-12	SHIPPED	2026-04-17T14:44:43.634+00:...	orders.csv
14	14	1493	2022-07-14	DELIVERED	2026-04-17T14:44:43.634+00:...	orders.csv
15	15	1420	2022-12-15	DELIVERED	2026-04-17T14:44:43.634+00:...	orders.csv

Orders Table



Silver Layer (Cleaning & Validation)



Silver Layer Tables

silver_customers

OverviewSample DataDetailsPermissionsPoliciesHistoryLineageInsightsQuality

Ask your question about the sample data...

NewAsk

Which countries have the most customer signups overall?How have customer signups trended by month over time?Are there customers sharing the same email address?

Sample

	customer_id	first_name	last_name	email	signup_date	country	ingestion_timestamp
1	1	Nikhil	Patel	nikhilpat@icloud.com	2020-03-06	India	2026-04-17T14:44:39.000Z
2	2	Ananya	Jackson	ananya_jackson@hotmail.com	2020-11-22	India	2026-04-17T14:44:39.000Z
3	3	Lisa	Reddy	lisa.reddy@yahoo.com	2020-09-12	United States	2026-04-17T14:44:39.000Z
4	4	Thomas	White	thomas.white@hotmail.com	2020-07-25	United States	2026-04-17T14:44:39.000Z
5	5	Sakshi	Brown	sbrown@protonmail.com	2021-05-16	United States	2026-04-17T14:44:39.000Z
6	6	Varun	Allen	varun.allen@protonmail.com	2023-03-07	India	2026-04-17T14:44:39.000Z
7	7	Charlotte	Walker	cwalker@zoho.com	2023-04-10	India	2026-04-17T14:44:39.000Z
8	8	Sebastian	Anderson	sebastian.anderson@yahoo.com	2022-02-27	Ireland	2026-04-17T14:44:39.000Z
9	9	Grace	Sharma	grace_sharma@yahoo.com	2020-01-07	Japan	2026-04-17T14:44:39.000Z
10	10	Abigail	Rao	arao@aol.com	2024-05-13	Singapore	2026-04-17T14:44:39.000Z
11	11	Evelyn	Patel	epatel@yahoo.com	2024-07-29	Australia	2026-04-17T14:44:39.000Z

Customers Table

silver_order_items

OverviewSample DataDetailsPermissionsPoliciesHistoryLineageInsightsQuality

Ask your question about the sample data...

NewAsk

Which products have the highest total ordered quantities?Are there orders with unusually high item quantities?How many order items are there per order?

Sample

	order_id	order_item_id	product_id	quantity	ingestion_timestamp	source_file_name
1	1787	3972	134	2	2026-04-17T14:44:45.683+00:00	order_items.csv
2	4773	10615	131	1	2026-04-17T14:44:45.683+00:00	order_items.csv
3	4209	9402	17	3	2026-04-17T14:44:45.683+00:00	order_items.csv
4	1767	3929	59	2	2026-04-17T14:44:45.683+00:00	order_items.csv
5	3026	6719	31	1	2026-04-17T14:44:45.683+00:00	order_items.csv
6	4725	10525	66	2	2026-04-17T14:44:45.683+00:00	order_items.csv
7	4392	9796	88	4	2026-04-17T14:44:45.683+00:00	order_items.csv
8	3359	7500	136	2	2026-04-17T14:44:45.683+00:00	order_items.csv
9	4593	10252	3	1	2026-04-17T14:44:45.683+00:00	order_items.csv
10	4462	9954	87	1	2026-04-17T14:44:45.683+00:00	order_items.csv
11	4162	9315	85	8	2026-04-17T14:44:45.683+00:00	order_items.csv
12	3510	7826	53	3	2026-04-17T14:44:45.683+00:00	order_items.csv

Order Items Table

silver_products

OverviewSample DataDetailsPermissionsPoliciesHistoryLineageInsightsQuality

Ask your question about the sample data...

NewAsk

Which categories have the highest average product price?How many products are missing names or categories?Are there price differences across categories?

Sample

	product_id	product_name	category	price	ingestion_timestamp	source_file_name
1	1	Pro Mouse	Electronics	77.44	2026-04-17T14:44:41.438+00:00	products.csv
2	2	HD Monitor Light	Electronics	36.42	2026-04-17T14:44:41.438+00:00	products.csv
3	3	Bluetooth Headphones	Electronics	225.44	2026-04-17T14:44:41.438+00:00	products.csv
4	4	Bluetooth Mouse	Electronics	263.59	2026-04-17T14:44:41.438+00:00	products.csv
5	5	Pro Camera	Electronics	285.31	2026-04-17T14:44:41.438+00:00	products.csv
6	6	Pro Speaker	Electronics	159.32	2026-04-17T14:44:41.438+00:00	products.csv
7	7	HD Speaker	Electronics	44.88	2026-04-17T14:44:41.438+00:00	products.csv
8	8	Wireless Monitor Light	Electronics	268.61	2026-04-17T14:44:41.438+00:00	products.csv
9	9	Smart Headphones	Electronics	10.22	2026-04-17T14:44:41.438+00:00	products.csv
10	10	Wireless Charger	Electronics	101.16	2026-04-17T14:44:41.438+00:00	products.csv
11	11	Pro Camera	Electronics	131.8	2026-04-17T14:44:41.438+00:00	products.csv
12	12	Bluetooth Speaker	Electronics	210.88	2026-04-17T14:44:41.438+00:00	products.csv

Products Table

silver_orders

OverviewSample DataDetailsPermissionsPoliciesHistoryLineageInsightsQuality

Ask your question about the sample data...

NewAsk

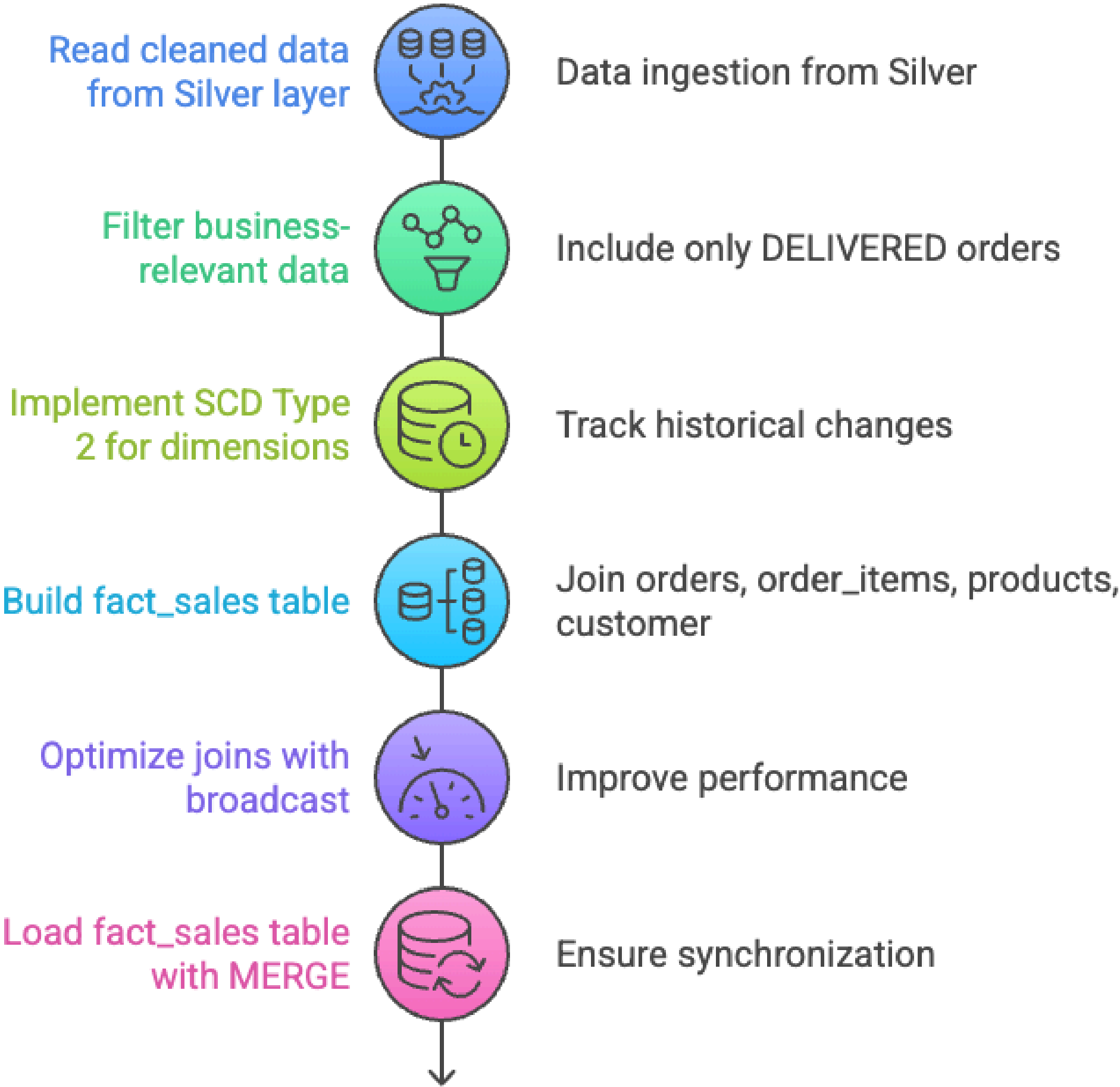
What is the most common order status?How many orders were placed by each customer?What is the trend of orders over time?

Sample

	order_id	customer_id	order_date	status	ingestion_timestamp	source_file_name
1	707	1583	2022-05-26	CANCELLED	2026-04-17T14:44:43.634+00:00	orders.csv
2	1394	1836	2022-05-26	CANCELLED	2026-04-17T14:44:43.634+00:00	orders.csv
3	2519	1043	2022-05-26	CREATED	2026-04-17T14:44:43.634+00:00	orders.csv
4	3242	1240	2022-05-26	DELIVERED	2026-04-17T14:44:43.634+00:00	orders.csv
5	4028	1494	2022-05-26	CREATED	2026-04-17T14:44:43.634+00:00	orders.csv
6	4742	1995	2022-05-26	DELIVERED	2026-04-17T14:44:43.634+00:00	orders.csv
7	4990	1556	2023-08-03	DELIVERED	2026-04-17T14:44:43.634+00:00	orders.csv
8	4153	1035	2023-08-03	SHIPPED	2026-04-17T14:44:43.634+00:00	orders.csv
9	4047	579	2023-08-03	SHIPPED	2026-04-17T14:44:43.634+00:00	orders.csv
10	3351	1115	2023-08-03	SHIPPED	2026-04-17T14:44:43.634+00:00	orders.csv
11	1861	5	2023-08-03	SHIPPED	2026-04-17T14:44:43.634+00:00	orders.csv
12	1068	290	2023-08-03	DELIVERED	2026-04-17T14:44:43.634+00:00	orders.csv
13	535	631	2023-08-03	SHIPPED	2026-04-17T14:44:43.634+00:00	orders.csv

Orders Table

Gold Layer (Data Modelling)



Gold Layer Tables

i_3^2 customer_id	A_0^B full_name	A_0^B email	A_0^B country	📅 signup_date	A_0^B hash_val
1	Nikhil Patel	nikhilpat@icloud.com	India	2020-03-06	42c060b8844e943bc5d002222db94886acf88d74eb294066e90ad2a9227
2	Ananya Jackson	ananya_jackson@hotmail.com	India	2020-11-22	32e18cf00cfacfa68deaf0d527071f07e0ecf138132854a3d5ffc3be5070fd8
3	Lisa Reddy	lisa.reddy@yahoo.com	United States	2020-09-12	397eaea71b37c3ef2a4ed2e61d7054339fea0790d56a6c4970c3c10bf779
4	Thomas White	thomas.white@hotmail.com	United States	2020-07-25	3f0ba0f24880202accdcd294b34bb9c55f8b228ddd7c2e722292adfb4332
5	Sakshi Brown	sbrown@protonmail.com	United States	2021-05-16	7cc50f588e824d6fb30b20945f240a6dc68b61aaa6e83b2eb12592dec042
6	Varun Allen	varun.allen@protonmail.com	India	2023-03-07	f9389ed175e56e418245e17af7ada38059692d9ce989ed50683389e9d8d
7	Charlotte Walker	cwalker@zoho.com	India	2023-04-10	7a5ab284c16f0e6107283c64c20e34f65c4d65d28fc4ae72be2f3c9326c7d
8	Sebastian Anderson	sebastian.anderson@yahoo.com	Ireland	2022-02-27	b6ebba6744300bca0a41e0842d5eac8ef2f350f77edd0a67f9cbe63d90bc
9	Grace Sharma	grace_sharma@yahoo.com	Japan	2020-01-07	d72cfd721445e99780afd70402a65a15022ec53fb63d98c64981aa2b0f31
10	Abigail Rao	arao@aol.com	Singapore	2024-05-13	02441c807a249dfa2ffbba649ddceb5f052837b9cc97f93c7a6ec6b79326e
11	Evelyn Patel	epatel@yahoo.com	Australia	2024-07-29	e9600dc3fabafc8b7edd4e5ad71667165b83265e8e6643194d56ffe08595
12	Priya Hall	priyahal@zoho.com	Canada	2024-02-03	28f3e7ed77c027d685c4e19fe85aa993378dcbafe2025bd8c39ab162d2e5
13	Betty Rodriguez	betty.rodriguez@aol.com	India	2021-04-13	8aa803c060a67e1209c4d9d513379f6f553be59c9e1b1cd0ce21d6e92b4f
14	Sebastian Pillai	sebastian.pillai86@icloud.com	United States	2024-05-25	565ae6e1ccd8fc30d8ee267e0afee095ca5ffc8d8f24d6ed4bf81fdda6911
15	Lucas Williams	lucas.williams53@mail.com	France	2024-01-31	5954a4cd7f0e19674956eea73ffa6b4231e18c50e2abfb649a77444853

i_3^2 product_id	A_0^B product_name	A_0^B category	A_0^B hash_val	📅 effective_from	📅 effective_to
1	Pro Mouse	Electronics	dd6a5859334a4bede5c22ee8076022a3b4d38c6325c7510455b4cfd3e58a2e1b	2026-04-18T13:45:22.802+00:...	null
2	HD Monitor Light	Electronics	de7a7b957b2b760a47009b6d2180a2169e504488795669fd77422cb29fffa30e	2026-04-18T13:45:22.802+00:...	null
3	Bluetooth Headphones	Electronics	0d83581d0102874709e4b02fa17a44ce1a384d5c8f1d0812354938215fd1edea	2026-04-18T13:45:22.802+00:...	null
4	Bluetooth Mouse	Electronics	f9e66a01aa433d252150a3a661dc7ed4f7dc477d754d3da76eeb391b25d6d473	2026-04-18T13:45:22.802+00:...	null
5	Pro Camera	Electronics	3d75f4547d5295e857755e01b847743f5f22f1e6aed22853834f5b71d3d52de0	2026-04-18T13:45:22.802+00:...	null
6	Pro Speaker	Electronics	f8c4c6c0dfde518acd4c727ddb1962cef0a46d7492037ed0e8e4962b9d33db3	2026-04-18T13:45:22.802+00:...	null
7	HD Speaker	Electronics	082b7eb9d79706deb8f28d67118e0777b75fa793fafd6a409c4a13817640fd87	2026-04-18T13:45:22.802+00:...	null
8	Wireless Monitor Light	Electronics	f541a79f91bdd7abb405e8d223b99951a2ae3c96cdb1bad61971158ecd315fd8	2026-04-18T13:45:22.802+00:...	null
9	Smart Headphones	Electronics	6e0b7e50e37d5336e0196222287e9df5226f472892faefa5adf536ed212444b0	2026-04-18T13:45:22.802+00:...	null
10	Wireless Charger	Electronics	89198df9279b0f59dda65167cf2414858ffa652ecc730155767bfe1ec9d71f15	2026-04-18T13:45:22.802+00:...	null
11	Pro Camera	Electronics	3d75f4547d5295e857755e01b847743f5f22f1e6aed22853834f5b71d3d52de0	2026-04-18T13:45:22.802+00:...	null
12	Bluetooth Speaker	Electronics	d96050c4f202459e0e0d45dc74c131a00ad298ba75b8aa71190b1f9d3698aeeb	2026-04-18T13:45:22.802+00:...	null
13	Portable Headphones	Electronics	77feb9f7f7807b939c636965d1c92272f94a5566e51289c514f2813d37f89047	2026-04-18T13:45:22.802+00:...	null
14	Classic Sweater	Clothing	6dcef0c96a916e6da37d8632e00c2151bb3b873351d6968167844025c98a4069	2026-04-18T13:45:22.802+00:...	null
15	Premium Shirt	Clothing	0df6a02a82eeef2e8d723887edb2c5160747e5d9f75a61f3ed6dba97e3bfdc65	2026-04-18T13:45:22.802+00:...	null

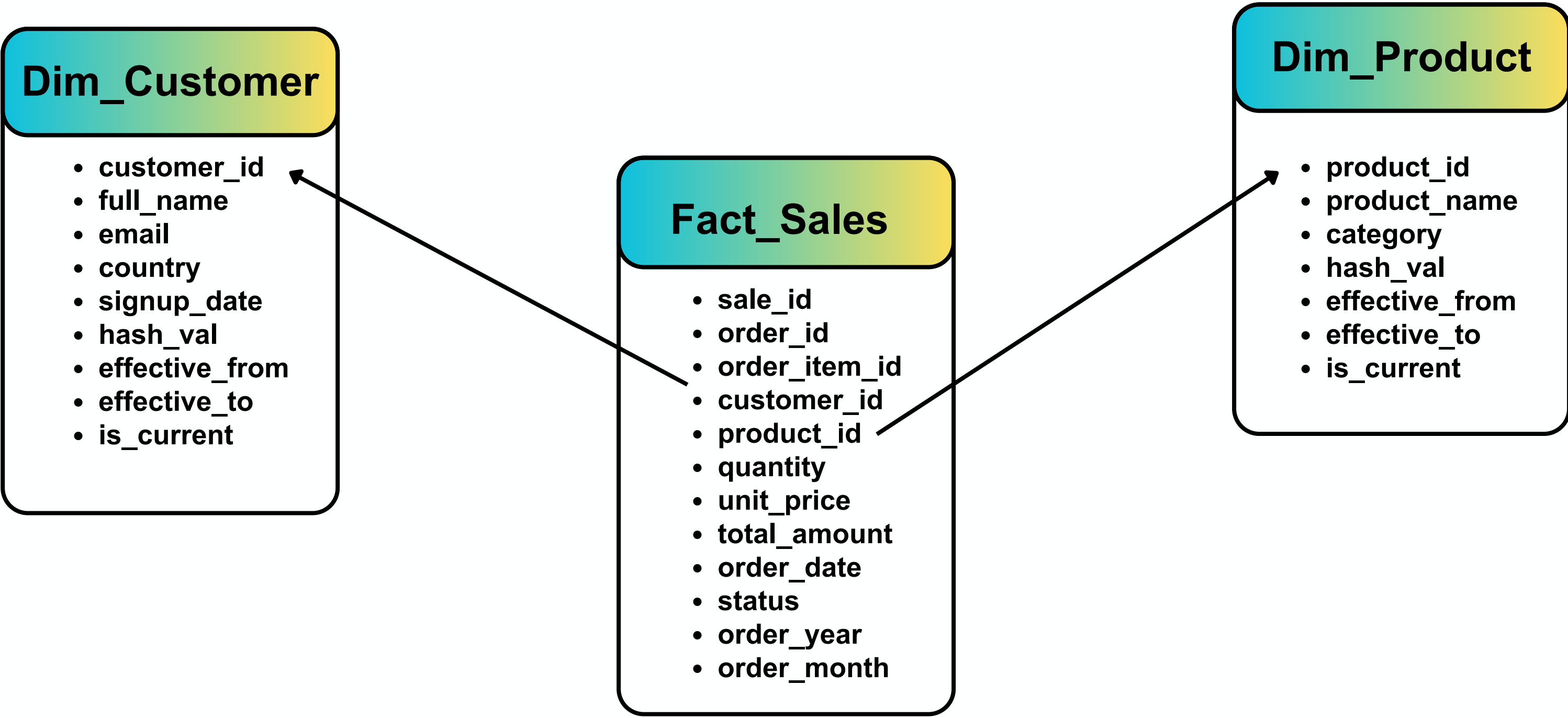
Customer Dimension

	A_0^B sale_id	i_3^2 order_id	i_3^2 order_item_id	i_3^2 customer_id	i_3^2 product_id	i_3^2 quantity	1.2 unit_price	1.2 total_amount	📅 order_date	A_0^B status	i_3^2 order_year
1	5f5f0e5d0fab762e1729d2d0fe5260030300a285df2815f2d48c8a904c6f8c44	54	122	1456	62	1	73.69	73.69	2025-03-08	DELIVERED	2025
2	008c1aab2c39224a6253d20c5def259c929779d269c71130dc608df033dfb5b0	3734	8354	1423	70	7	23.95	167.65	2025-03-08	DELIVERED	2025
3	a65102c39f852e9db14bc607da4dd52034f08730737e89c9ca73f9a75adcae63	3576	7979	230	40	8	21.98	175.84	2025-03-08	DELIVERED	2025
4	290cdad2231fbb8b4cee4752888f6ed5ca3a9ede49b04707adb82686b362741a	3734	8350	1423	95	5	29.54	147.7	2025-03-08	DELIVERED	2025
5	7f47c94b80f80a703ede8913da01ceb85f49da335a531030a00e8ccf9663ff9a	3734	8351	1423	137	7	40.19	281.33	2025-03-08	DELIVERED	2025
6	7bf3dccde4d1534daac525143415ce7fefb3069acdb96ee9fb97931c0ab7b096	3734	8353	1423	104	3	40.42	121.26	2025-03-08	DELIVERED	2025
7	32c09ebdf6468a656acc4251413f9eb0bac6e5633a3b78adc916f65bb5a86ef2	54	121	1456	39	10	103.22	1032.2	2025-03-08	DELIVERED	2025
8	0ac0f0e2f6c850b574c86d2360154c1d3050f863b4055ef1f4fdbde602383d6e	3576	7978	230	41	2	35.2	70.4	2025-03-08	DELIVERED	2025
9	d33e8418ad00cb42967416d4b188f0842aeaa552a1bd9e996b7385100436bfec	3734	8352	1423	10	5	101.16	505.8	2025-03-08	DELIVERED	2025
10	2225b8ec9e55f9c163cf321c5c89726fb7c892a7ebcb1cbf25a1b84e1610d9d	54	119	1456	101	1	7.37	7.37	2025-03-08	DELIVERED	2025
11	72d26955bb578c0fbd77856f5fdc5e3f754683dbeb66b8bc41de490b733108e9	54	120	1456	56	2	102.05	204.1	2025-03-08	DELIVERED	2025
12	e937d26d616b55c1ef11ea9cafbae01720ac72065a5deb9d0d504854ed287eba	1092	2424	1962	8	2	268.61	537.22	2024-06-23	DELIVERED	2024
13	6118a4095d50bc0254e30e0de09e31463f2dd14f47201f2feb570c3a851fe422	3189	7101	474	84	9	82.31	740.79	2024-06-23	DELIVERED	2024
14	d78c870e55f2fb5c5bad747bb1dc2fcd345e05e3c51c30e302fa349ea39ec32f	2826	6277	1562	84	2	82.31	164.62	2024-06-23	DELIVERED	2024
15	71426c1a42eb5b6402036b2aa77493b7f7ceb6a155e3ff0e710c977db7d01255	3189	7100	474	27	3	189.2	567.6	2024-06-23	DELIVERED	2024
16	89d3fbc17e74afb2fdae393b47fad11436a7b10069e8e4b0395978ef7dca96ab	2826	6278	1562	146	4	63.04	252.16	2024-06-23	DELIVERED	2024
17	d19c5ea54fbe5fc3160b28abc0293bf64cfba0b2bd81cf2ad95528930bb95c6d	3189	7102	474	31	2	48.22	96.44	2024-06-23	DELIVERED	2024
18	b5fd3afb0ab9d8ee25fbb6349f53d5afe3c53af223e494ae81a2dcff0cddde4e7	4411	9841	1404	25	1	91.04	91.04	2024-06-23	DELIVERED	2024

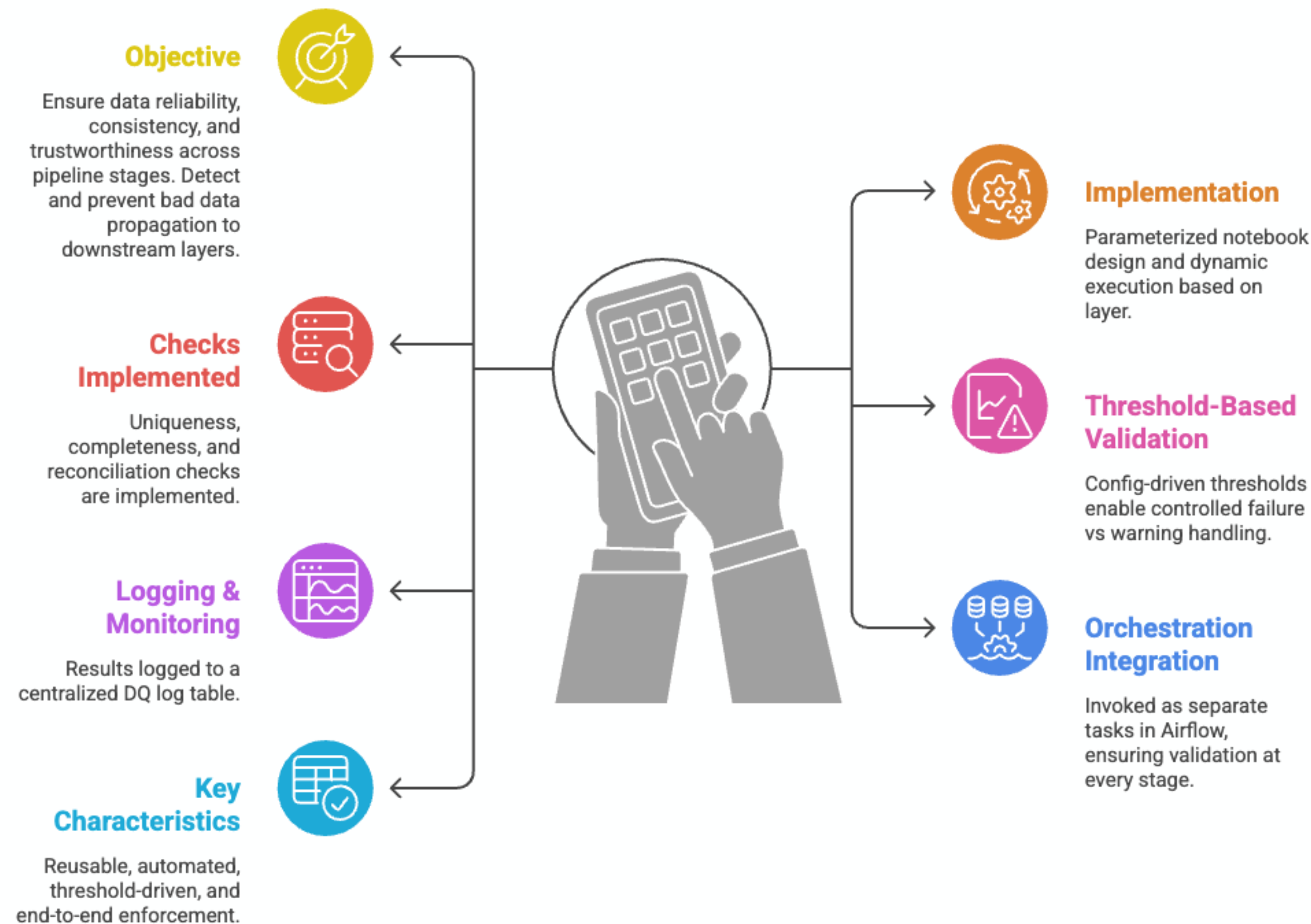
Fact Sales



Star Schema



Data Quality Framework



Data Quality Logs Table

Catalog Explorer > capstone_project_catalog > logs >

dq_logs

OverviewSample DataDetailsPermissionsPoliciesHistoryLineageInsightsQuality

Ask your question about the sample data...

Which tables generate the most failed data quality checks?

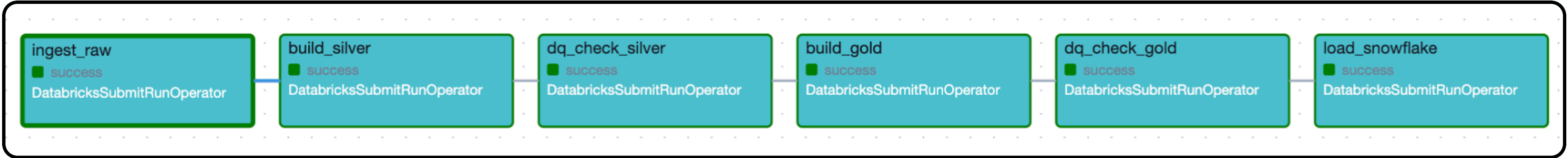
How do data quality statuses trend over time?

Which columns most frequently appear in failed checks?

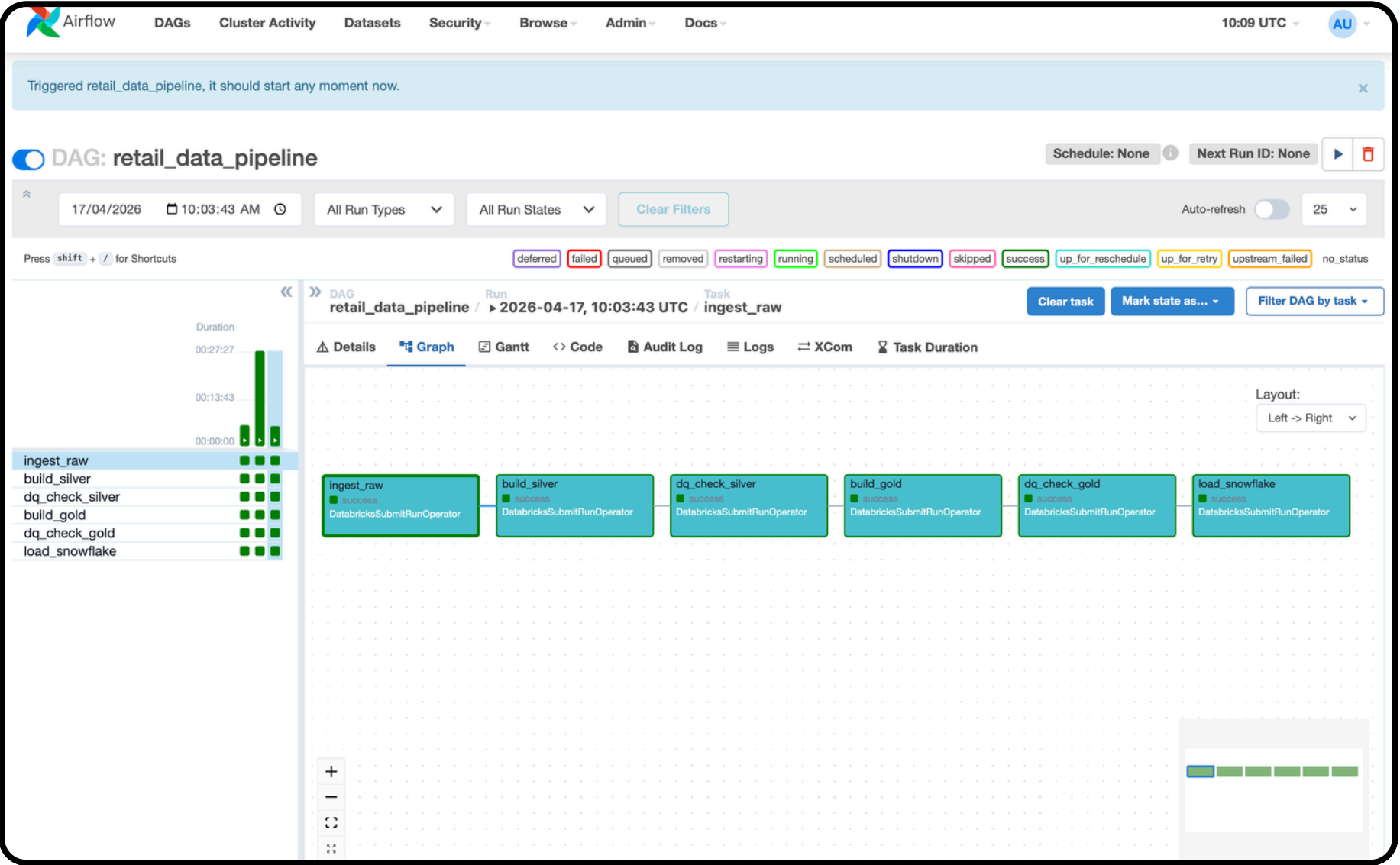
Sample

	check_type	table	column	status	details	timestamp
1	UNIQUENESS	capstone_project_catalog.gold.fact_sales	sale_id	PASS	{'total': 5240, 'distinct': 5240, 'dups': ...}	2026-04-17T14:18:01.8999...
2	UNIQUENESS	capstone_project_catalog.gold.dim_custom...	customer_id	PASS	{'total': 1947, 'distinct': 1947, 'dups': ...}	2026-04-17T14:18:02.4767...
3	UNIQUENESS	capstone_project_catalog.gold.dim_product	product_id	PASS	{'total': 149, 'distinct': 149, 'dups': 0}	2026-04-17T14:18:03.0920...
4	COMPLETENESS	capstone_project_catalog.gold.fact_sales	order_id	PASS	{'nulls': 0, 'pct': 0.0}	2026-04-17T14:18:04.0529...
5	COMPLETENESS	capstone_project_catalog.gold.fact_sales	customer_id	PASS	{'nulls': 0, 'pct': 0.0}	2026-04-17T14:18:04.6057...
6	COMPLETENESS	capstone_project_catalog.gold.fact_sales	product_id	PASS	{'nulls': 0, 'pct': 0.0}	2026-04-17T14:18:05.1553...
7	COMPLETENESS	capstone_project_catalog.gold.fact_sales	order_date	PASS	{'nulls': 0, 'pct': 0.0}	2026-04-17T14:18:05.4361...
8	COMPLETENESS	capstone_project_catalog.gold.fact_sales	total_amount	PASS	{'nulls': 0, 'pct': 0.0}	2026-04-17T14:18:05.9854...
9	COMPLETENESS	capstone_project_catalog.gold.fact_sales	quantity	PASS	{'nulls': 0, 'pct': 0.0}	2026-04-17T14:18:06.5101...
10	COMPLETENESS	capstone_project_catalog.gold.fact_sales	unit_price	PASS	{'nulls': 0, 'pct': 0.0}	2026-04-17T14:18:07.1293...
11	COMPLETENESS	capstone_project_catalog.gold.dim_custom...	customer_id	PASS	{'nulls': 0, 'pct': 0.0}	2026-04-17T14:18:07.7234...
12	COMPLETENESS	capstone_project_catalog.gold.dim_custom...	full_name	PASS	{'nulls': 0, 'pct': 0.0}	2026-04-17T14:18:08.0400...

Airflow DAG



Orchestration using Airflow

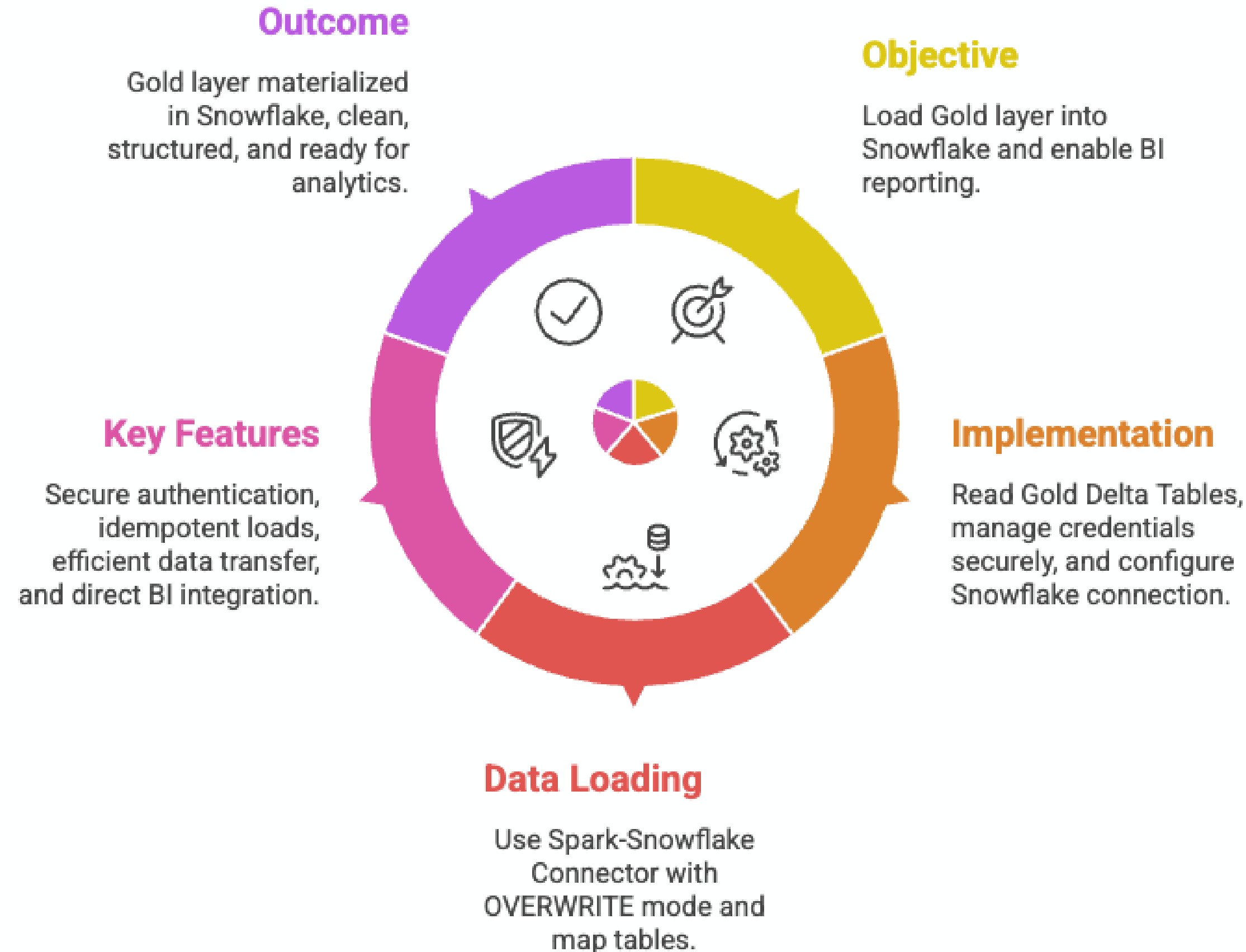


Sucessful DAG Run On Airflow UI

Alerts	Apr 17, 2026, 03:38 PM	load_snowflake	Job	ANURAG SA...	By runs submit API	7s	Succeed...
Query History	Apr 17, 2026, 03:38 PM	dq_check_gold	Job	ANURAG SA...	By runs submit API	23s	Succeed...
SQL Warehouses	Apr 17, 2026, 03:36 PM	build_gold	Job	ANURAG SA...	By runs submit API	1m 9s	Succeed...
Data Engineering	Apr 17, 2026, 03:36 PM	dq_check_silver	Job	ANURAG SA...	By runs submit API	12s	Succeed...
Runs	Apr 17, 2026, 03:34 PM	build_silver	Job	ANURAG SA...	By runs submit API	1m 14s	Succeed...
Data Ingestion	Apr 17, 2026, 03:33 PM	ingest_raw	Job	ANURAG SA...	By runs submit API	24s	Succeed...

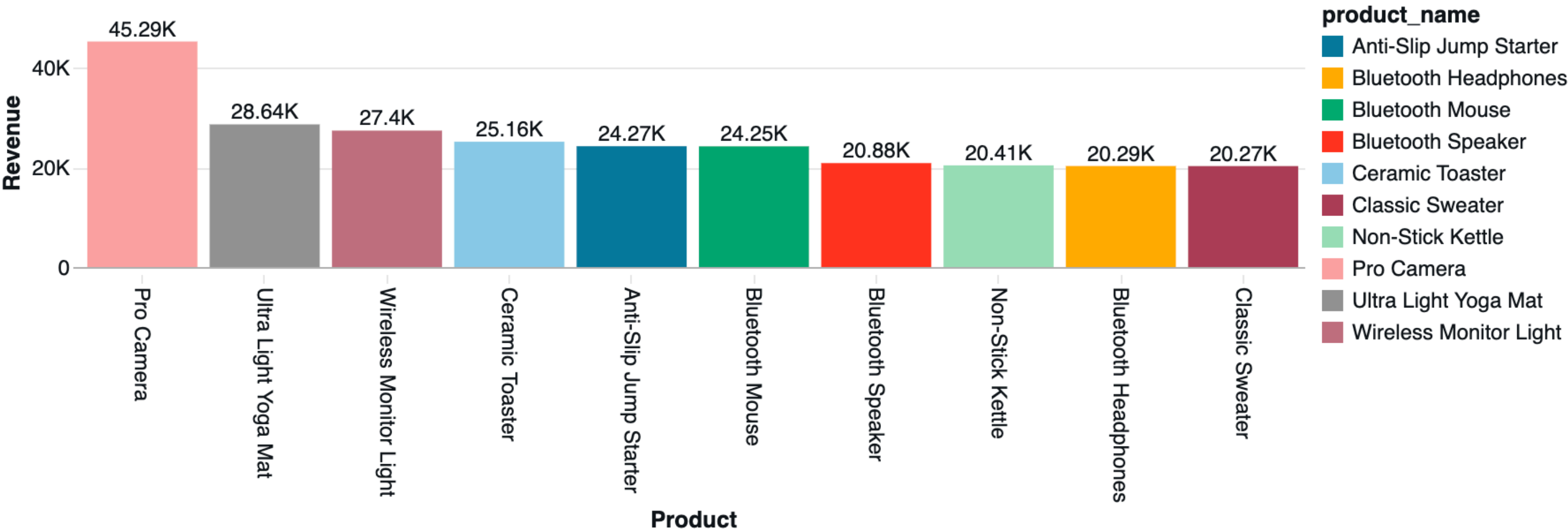
Successful Job Run on Databricks UI

Snowflake Integration

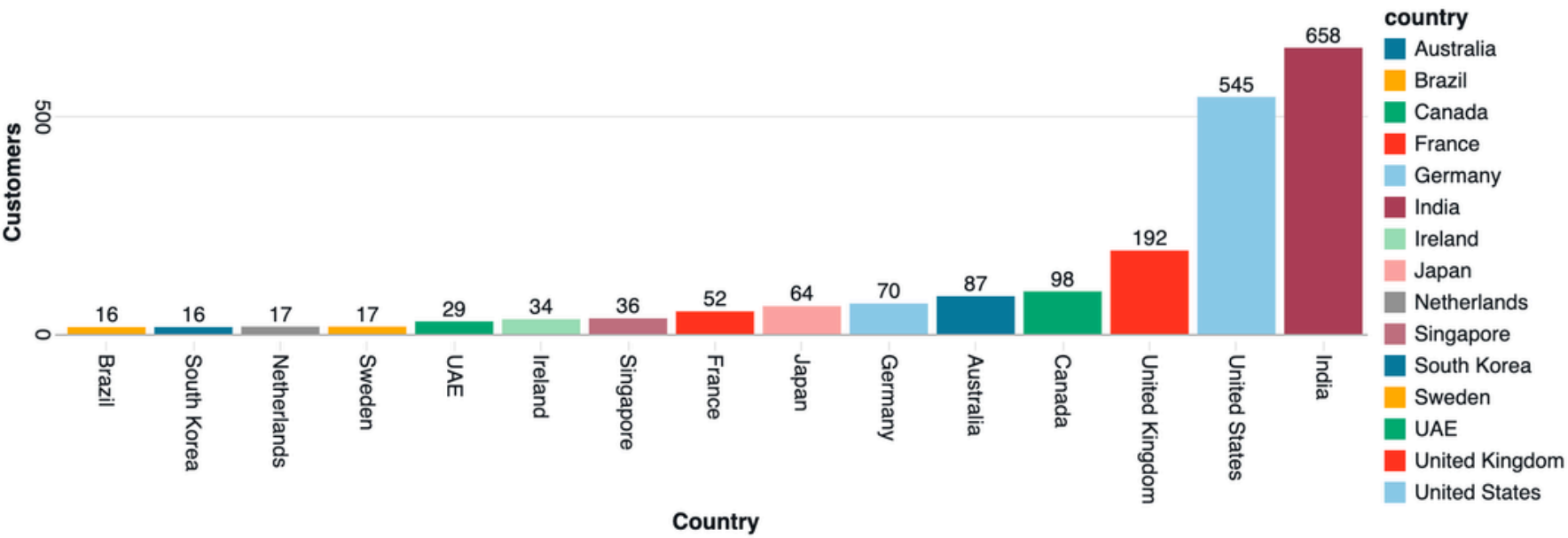


Business Insights

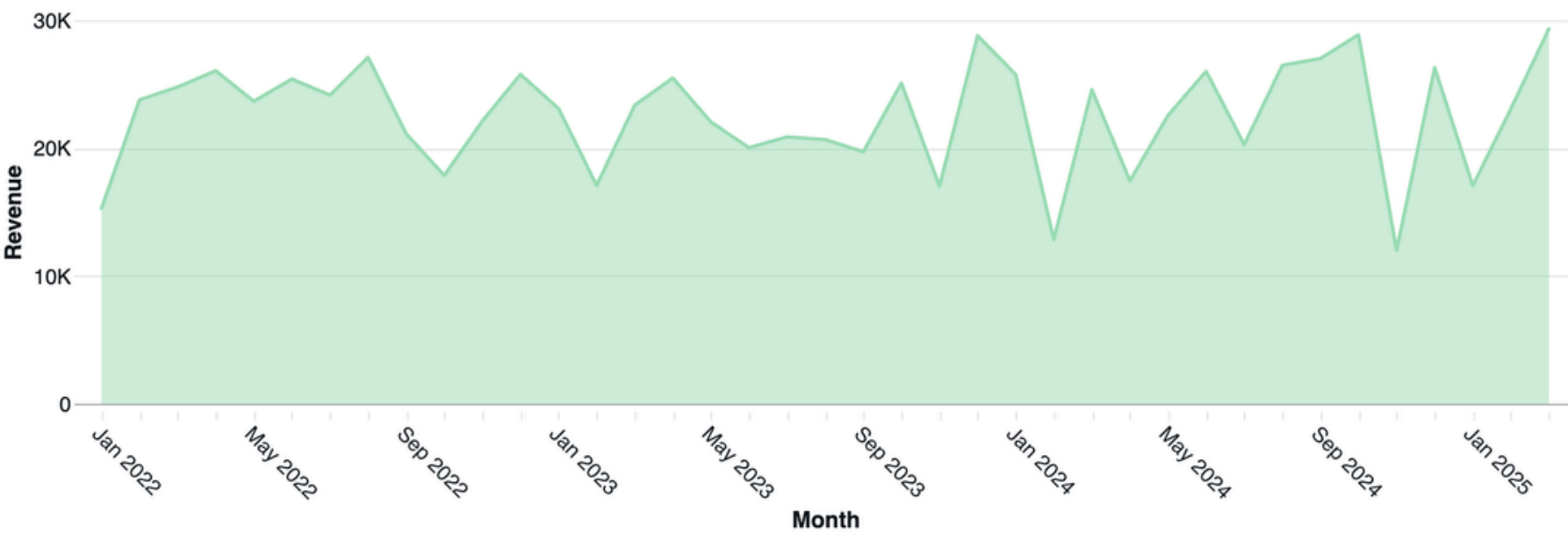
Top 10 Products by Revenue



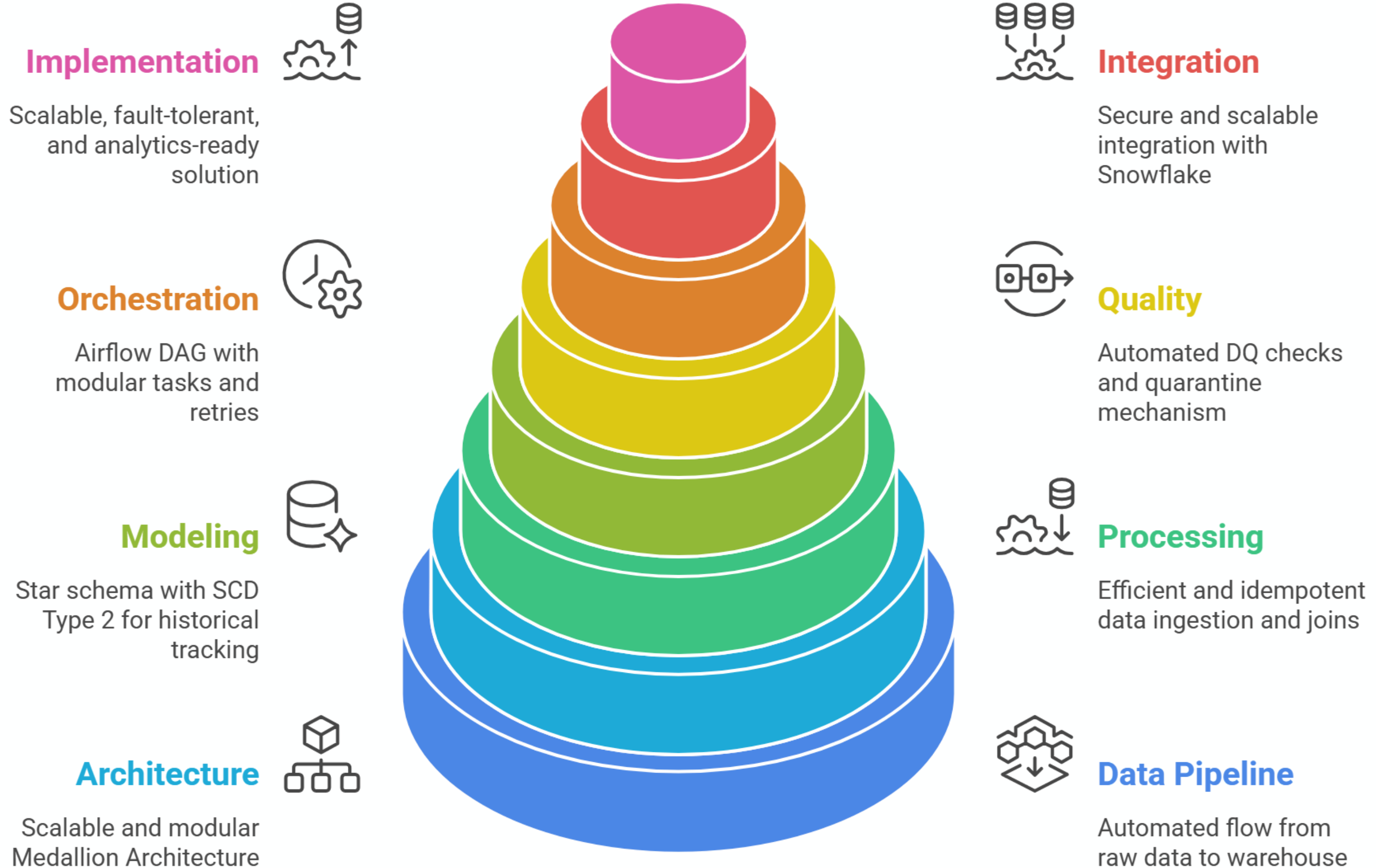
Customer Geographic Distribution



Revenue Performance Trend



What We Achieved



Challenges & Engineering Decisions

Challenge	Decision
 Dirty & Inconsistent Data	Permissive mode + corrupt record capture
 Incremental vs Full Load Processing	Incremental pipeline using ingestion_timestamp
 Maintaining Data Integrity	Integrity checks (foreign key validation)
 Handling Slowly Changing Data	SCD Type 2 (MERGE-based approach)
 Idempotent & Reliable Pipelines	MERGE (upsert) logic across Silver & Gold
 Reusability & Modularity	Config-driven pipeline (config.yaml)
 Performance Optimization	Broadcast joins for small tables (products)
 Orchestration & Dependency Management	Airflow DAG with clear task dependencies
 Security & Credential Management	Databricks Secrets for Snowflake integration

Thank You